

Experiment – 1.4.1

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Subject: Full Stack-II

Subject Code: 24CSP-337

1. Aim: To design and implement an interactive calendar interface for scheduling and managing posts.

2. Software Required:

- React.js
- Calendar library (e.g., FullCalendar / custom)
- Date utility libraries (optional)
- Code Editor

3. Objective:

- To understand time-based data visualization in UI systems
- To implement calendar-based scheduling interfaces
- To map structured data to temporal layouts
- To enable user interactions such as drag-and-drop

4. Theory:

Calendars are essential UI components for representing time-based data. In applications such as social media schedulers, events (posts) must be mapped to specific dates and time slots.

A calendar system involves:

- Temporal data modeling (date, time, duration)
- Event mapping (linking posts to time slots)
- User interaction handling (click, drag, resize)

Interactive calendars enhance usability by allowing users to:

- Visually organize content
- Modify schedules dynamically
- Manage large datasets efficiently

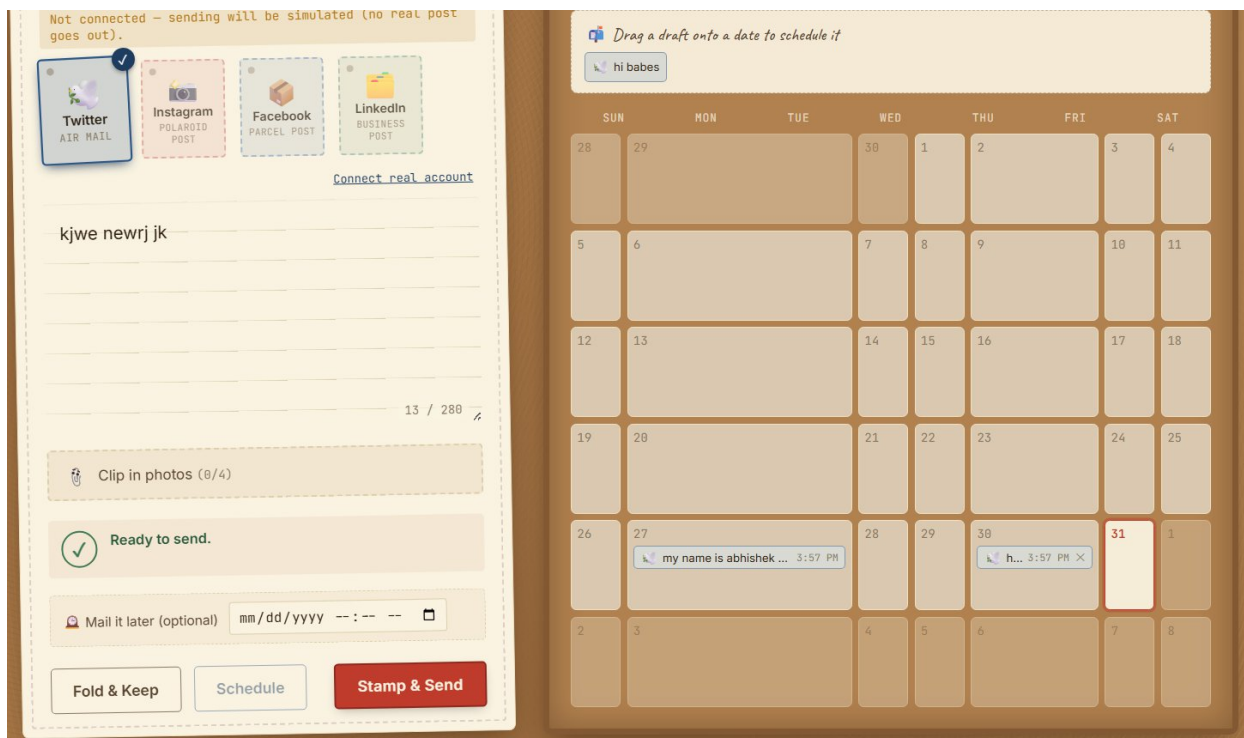
Academic Insight: Calendar systems fall under Human-Computer Interaction (HCI) and Information Visualization, where data is presented in a form that supports intuitive understanding and interaction.

5. Procedure/Algorithm:

- Design calendar UI layout (day/week/month view)
- Map post data to calendar events
- Render events dynamically
- Implement click interactions (view/edit)
- Add drag-and-drop scheduling (optional/advanced)
- Sync calendar with application state

6. Code and Output:

```
# SchedulingCalendar.css X
client > src > components > Calendar > # SchedulingCalendar.css > .sched-cal
1  .sched-cal {
2    background: color-mix(in srgb, var(--cork-1) 55%, var(--cork-2) 45%);
3    border: 6px solid color-mix(in srgb, var(--cork-3) 70%, #5a3c1f 30%);
4    border-radius: 8px;
5    padding: 20px;
6    box-shadow: inset 0 0 30px var(--cork-shadow), var(--shadow-lift);
7    min-height: 420px;
8  }
9
10 .sched-cal__header {
11   display: flex;
12   align-items: flex-start;
13   justify-content: space-between;
14   flex-wrap: wrap;
15   gap: 10px;
16   margin-bottom: 14px;
17 }
18
19 .sched-cal__header h3 {
20   font-size: 1.6rem;
21   color: var(--paper);
22   text-shadow: 0 2px 3px rgba(0, 0, 0, 0.35);
23   margin: 0;
24 }
25
26 .sched-cal__count {
27   font-family: var(--font-mono);
```



7. Conclusion:

- Functional calendar interface
- Posts mapped to time slots
- Interactive scheduling system
- Improved user experience for content planning

8. Learning Outcomes:

- Learned to design and implement an interactive calendar interface for scheduling posts
- Understood time-based data visualization and temporal layout mapping in UI systems
- Implemented event mapping to link post data with calendar time slots
- Gained experience handling user interactions such as click and drag-and-drop
- Enhanced skills in building dynamic, interactive scheduling systems in React